JIANXUAN SHAO

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RESEARCH INTERESTS

Human-AI Collaboration | Context-Aware Personal AI | Multimodal Interaction | Interaction Evaluation

Proposed doctoral research: Investigating how people and personal AI maintain shared context, turn dialogue into revisable actions, and adapt through repeated collaboration in everyday activities.

EDUCATION

University of Sydney Feb 2025 - Present

Master of Computer Science (Data Science and AI), expected Dec 2026

Average across 12 graded units: 79.33/100; Semester 1, 2026 average: 84.5/100. Selected results: Machine Learning and Data Mining 89 HD; Introduction to Programming 87 HD; Data Structures and Algorithms 77 DI; Introduction to Cybersecurity 92 HD; Computer and Network Organisation 85 HD.

Hebei University Sep 2018 - Jun 2023

Bachelor of Engineering in Architecture

GPA 4.05/5.0. Five-year degree covering architectural design, architectural physics, urban planning, computational design and digital visualisation.

SELECTED RESEARCH AND COMPUTATIONAL PROJECTS

Explainable AI-Generated Image Detection 2026 - Present

Research design and analysis lead | University of Sydney | Eight-student capstone team

- Led research framing, experimental planning and result analysis; implemented the direct detection baseline. Teammates implemented other branches, integrated fusion and conducted module-level evaluations.
- Combined two Qwen3.5-9B routes, verdict before explanation and evidence before verdict, with ResNet-50 visual forensics through logistic fusion. FakeBench evaluation used 6,000 images: 3,000 real and 3,000 fake from 10 generators.
- Team result: fusion achieved 94.33% balanced accuracy on the 900-image evaluation split, 5.67 percentage points above ResNet-50, the strongest single branch.
- Exploratory ablations refitted only the logistic fusion classifier on validation-set predictions. Removing ResNet-50 reduced evaluation balanced accuracy by 7.22 points; removing evidence-before-verdict reasoning reduced it by 0.44, with the latter difference not statistically confirmed.
- Analysed detection errors alongside generated explanations. Better text similarity did not yield statistically confirmed explanation-quality gains; explanations were assessed on 200 fake images using automatic metrics and a model judge.
- The language model was selected using scores that included this evaluation split; ablations were post hoc. External testing and human explanation evaluation remain future work.

AI Game Companion - Game Fellows Initiated 2025

Prototype completed | Prototype development and study preparation

- Developed a Python/Pygame co-play testbed by extending a Galaga-style game with a visible, rule-based teammate for threat interception, shielding and coordinated support fire.
- Linked text/audio callouts to support actions; used cooldowns and stable behaviour states to limit repeated prompts and rapid switching.
- Designed and implemented counterbalanced Solo/Fellow trials with paired random seeds and gameplay/event logs. Prepared questionnaires on presence, trust, usefulness and player control; no recruited-user study was conducted.

Machine Learning Classification and PathMNIST Pipelines 2026

COMP5318

- Built classification workflows with preprocessing, stratified cross-validation and hyperparameter tuning; extended the pipeline to PathMNIST using Random Forest, MLP and CNN baselines.
- Compared accuracy, macro-F1, confusion matrices and runtime trade-offs, with attention to class-level errors and model behaviour.

Electricity Sector Data Engineering and Spatial Integration 2026

COMP5339

• Integrated electricity, emissions, renewable-station and regional data with geocoded coordinates and spatial DuckDB storage. Processed 716,454 facility records and streamed JSON via MQTT to a Streamlit map dashboard.

BUILT ENVIRONMENT AND INTERACTIVE TECHNOLOGY EXPERIENCE

Shanghai Fengyuzhu Culture Technology Co., Ltd. Jul 2023 - Jul 2024

Interactive Media Intern

• Supported immersive spatial systems integrating CAVE projection, wall-touch interfaces, radar recognition and VR/AR visualisation, connecting architectural environments with interactive digital technologies.

• Worked across design, technical delivery and on-site system debugging, translating spatial concepts into user-facing interactive experiences.

Hebei Haoyu Architectural Design Co., Ltd. - Beidaihe Branch Jun 2022 - Aug 2022

Architectural Design Intern

• Contributed to an experimental primary-school project through site and user-needs research, spatial layout studies, client feedback iterations and design development.

• Produced floor plans, elevations, sections and visualisations in CAD-based architectural workflows.

COMPUTATIONAL DESIGN AND ARCHITECTURAL EXPERIENCE

Surveying and Digital Reconstruction of Daci Pavilion and Geyuan Temple Oct 2021

Team Member

• Recorded historic architectural geometry through sketches and CAD/Rhino documentation, reconstructing plans, elevations, sections and three-dimensional information for preservation.

• Modelled architectural components including traditional bracket forms; project records were archived in Baoding City Library.

Qinhuangdao Guangjian Construction Project Management Co., Ltd. Jun 2021 - Aug 2021

Intern / Assistant Architect

• Produced demolition, facade, floor-plan and sectional drawings for a shantytown redevelopment project, following senior architects' design requirements.

ADDITIONAL TECHNICAL AND INTERDISCIPLINARY EXPERIENCE

Mobile security analysis (INFO5995): Threat modelling and reproducible technical reporting of Android authentication logic.

Computational design training: Rhino/Grasshopper workshops, SCI-Arc digital visualisation workshop, UIA-HYP competitions (2021, 2022).

TECHNICAL SKILLS

Programming and prototyping: Python, SQL, interactive prototyping and data processing.

Research and evaluation: Reproducible model evaluation, error analysis and controlled study preparation.

Design and interaction: Computational and spatial design, 3D modelling and immersive interfaces.

AWARDS AND LEADERSHIP

Hebei University Outstanding Student Scholarship - Second Prize (2019); Third Prize (2021, 2022).

JYZC Undergraduate Innovation & Creativity Contest - Nomination Award (2022).

President, Hebei University Tennis Club (2019-2020); Team Leader, Hebei University architecture-themed campus sketching activity (2022).